



Carleton
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Pursuing the Visualization Leviathan: an Introduction to Making Visualizations Using LaTeX

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Introduction – Visualization

- I personally dislike most graphing software, either too difficult for what you get (Python's Plotly), or simple but still ugly unless you put in a lot of effort (Python's Matplotlib, Excel, MATLAB).

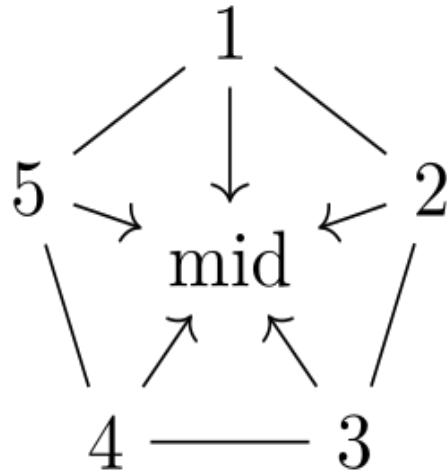
Highly Subjective Opinion on the Topic

Software Name	Ease of Use (1-5)	"Niceness" of Result (1-5)
Excel	5	1
Python - Matplotlib	4	3
OriginLab	4	4
MATLAB	4	3
Python - Seaborn	3	4
Python - Plotly	2.5	4
LaTeX	2	5
Rstudio (tidyverse)	1	5

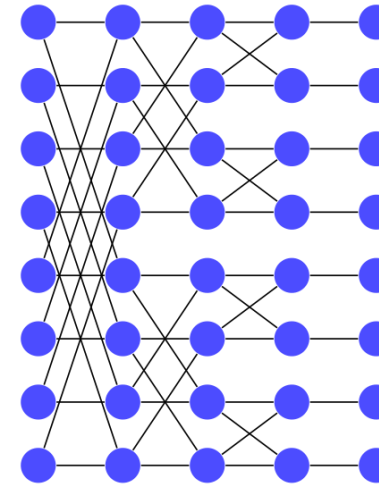
I Will Not Bother (Extensively) Defining LaTeX

-It's a typesetting program usually used for creating documents. Built on top of TeX (I don't think anyone interfaces directly with TeX anymore).
- It's pronounced lay-tek, **not** lay-tex. Rookie mistake!
- Today, we're introducing two particular packages of LaTeX, called:
 1. TikZ (built on top of PGF. Again, these details don't really matter 99.99% of the time)
 2. Pgfplots (which our tutorial will use)

- Useful for graphs in the mathematical sense (like, vertices and edges and that)



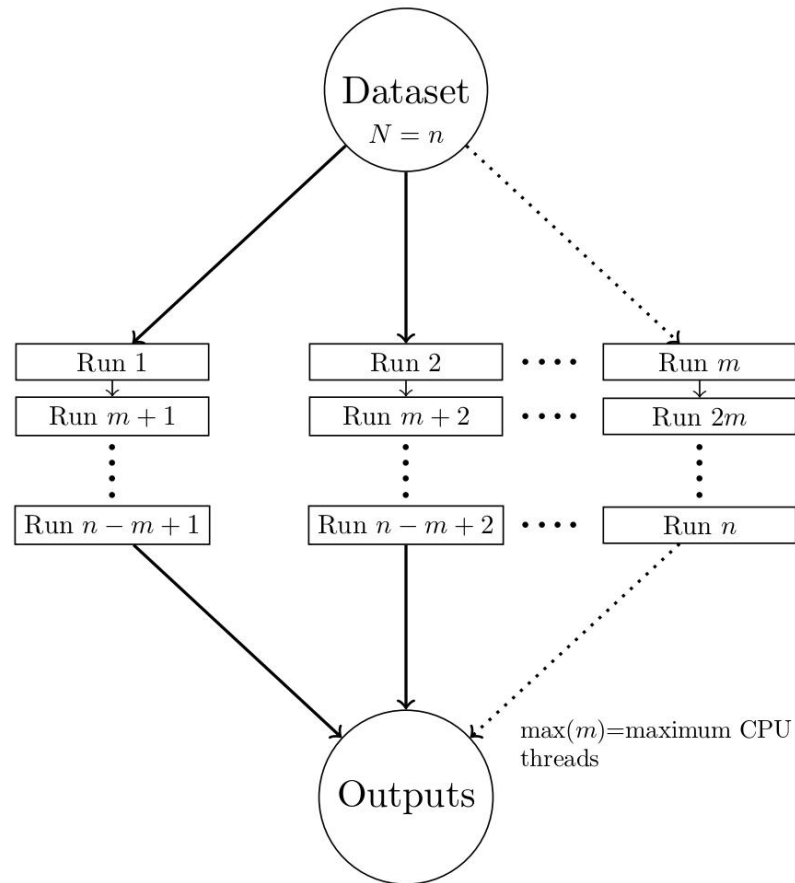
1 line of code



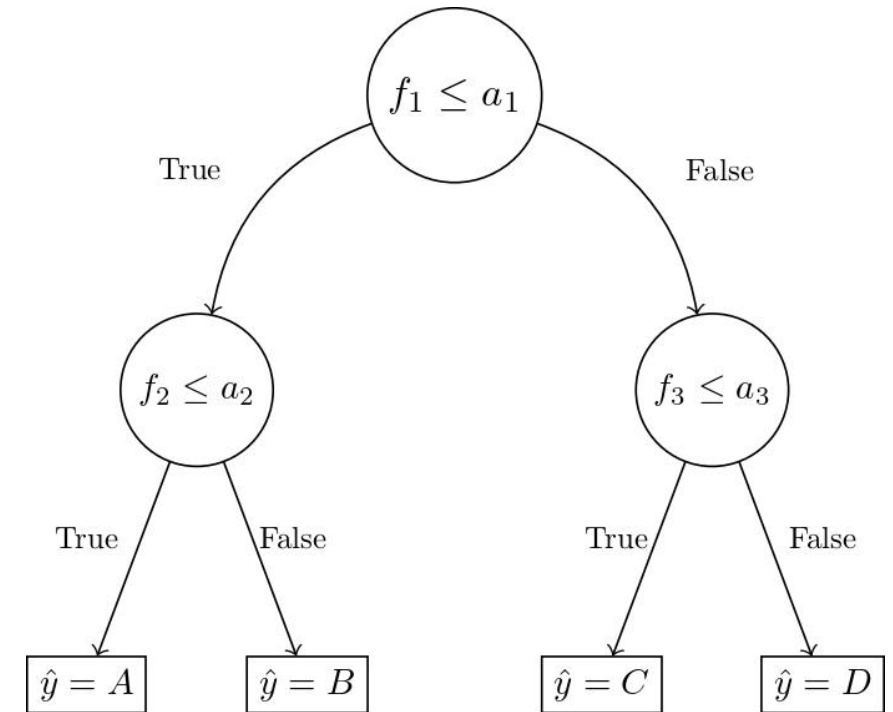
6 lines of code

Source: TikZ documentation (line count excludes preamble and start/end lines)

TikZ – Further Examples



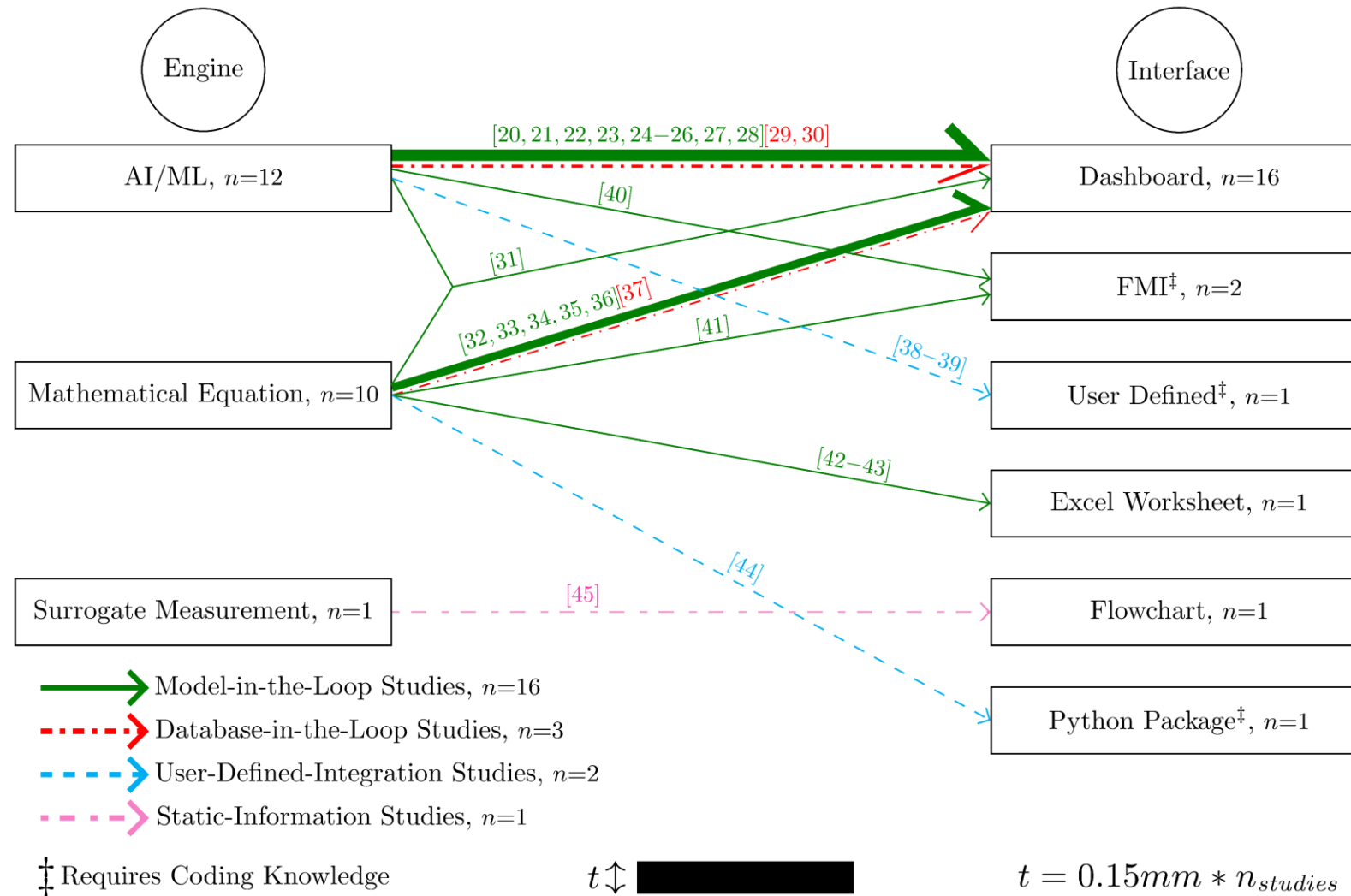
27 lines of code



17 lines of code

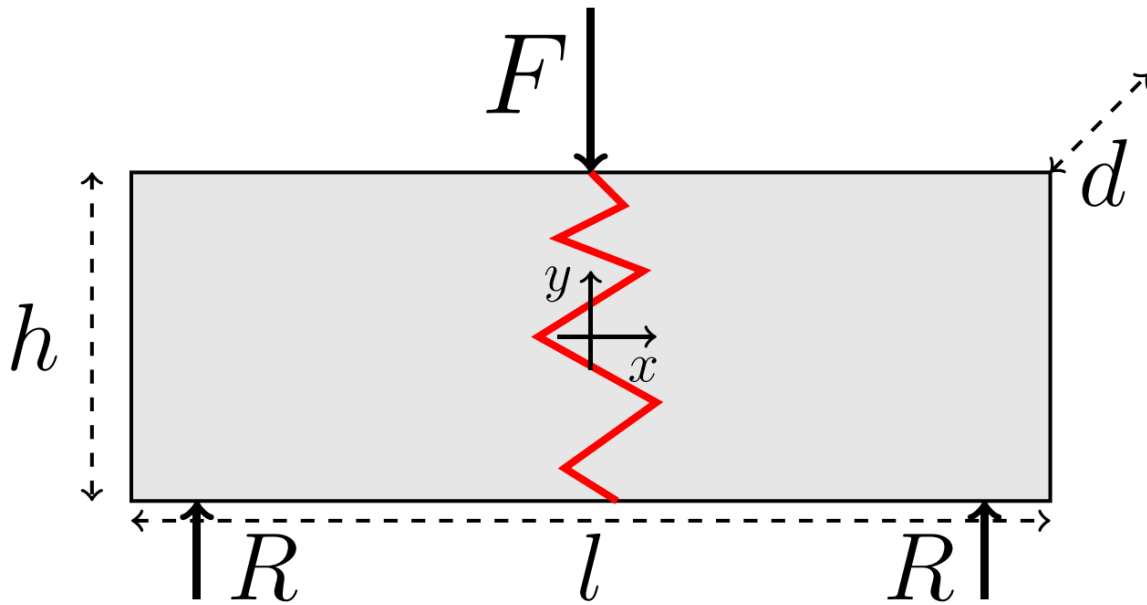
Source: Done inefficiently by me (same caveats as last slide)

TikZ – Scaling Problems if Done Inefficiently



49 lines of very dense code (also done by me)

TikZ – Further Examples



LaTeX, probably TikZ
From [A. Aryanfar et al. \(2024\)](#)

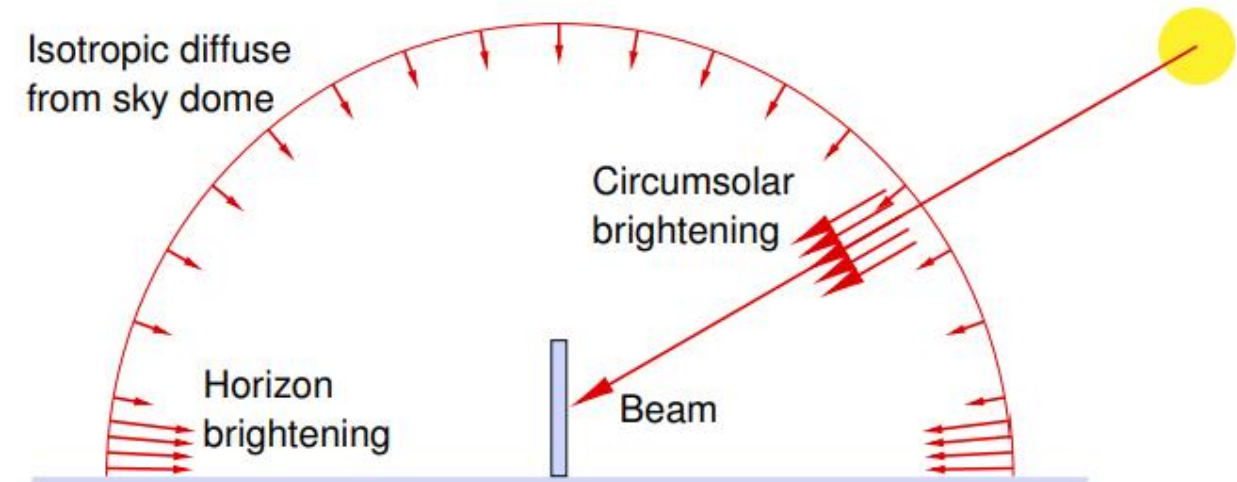
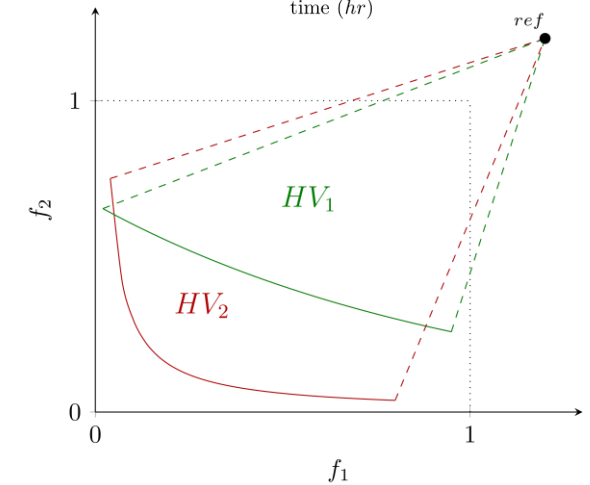
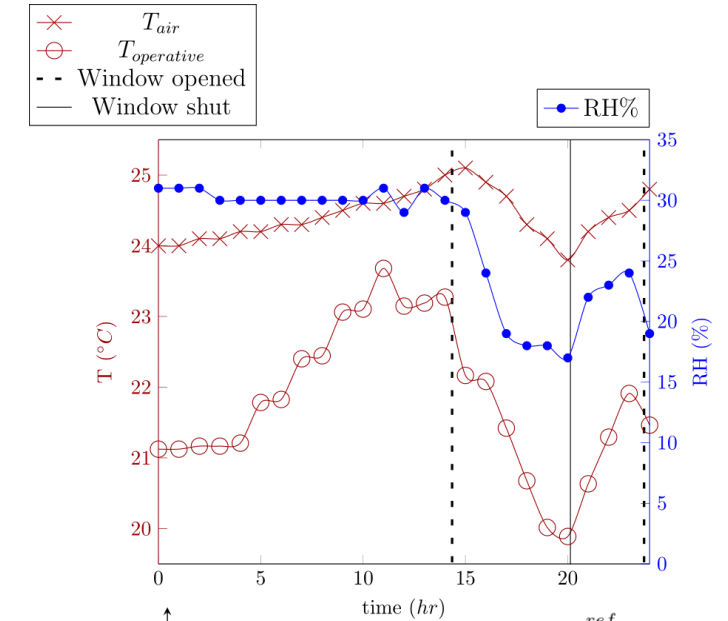
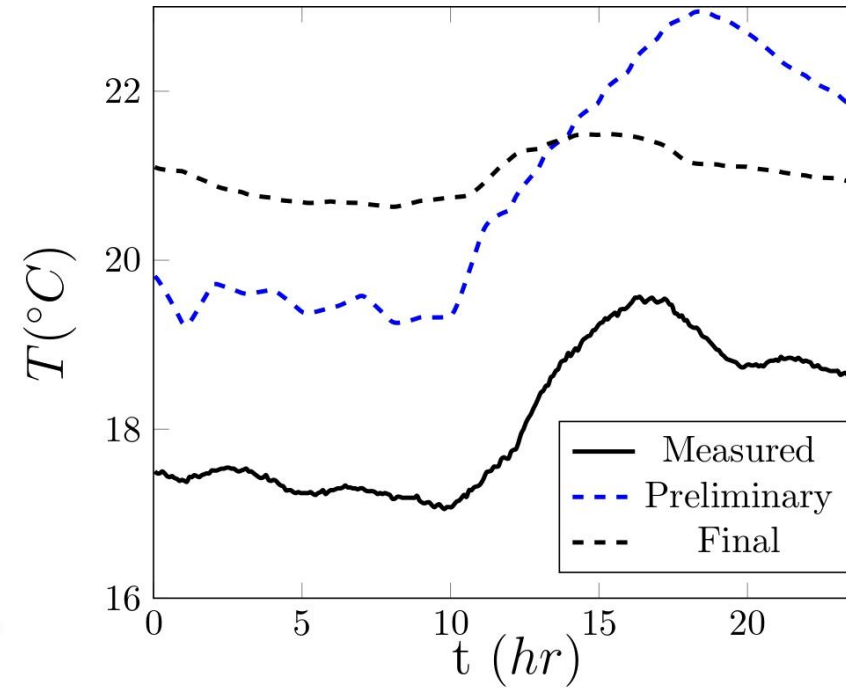
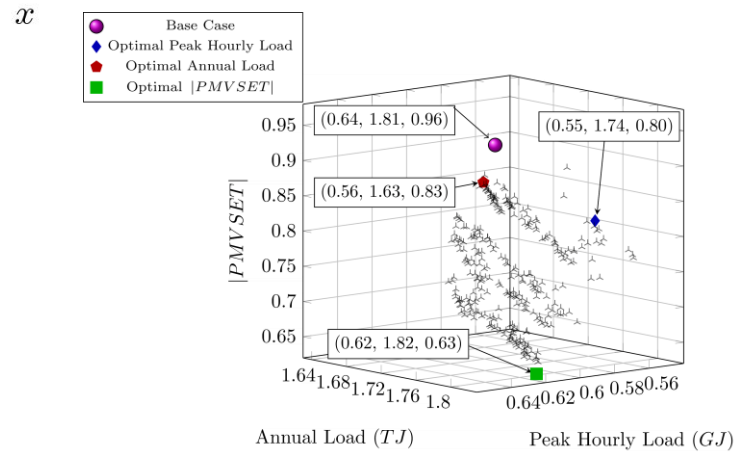
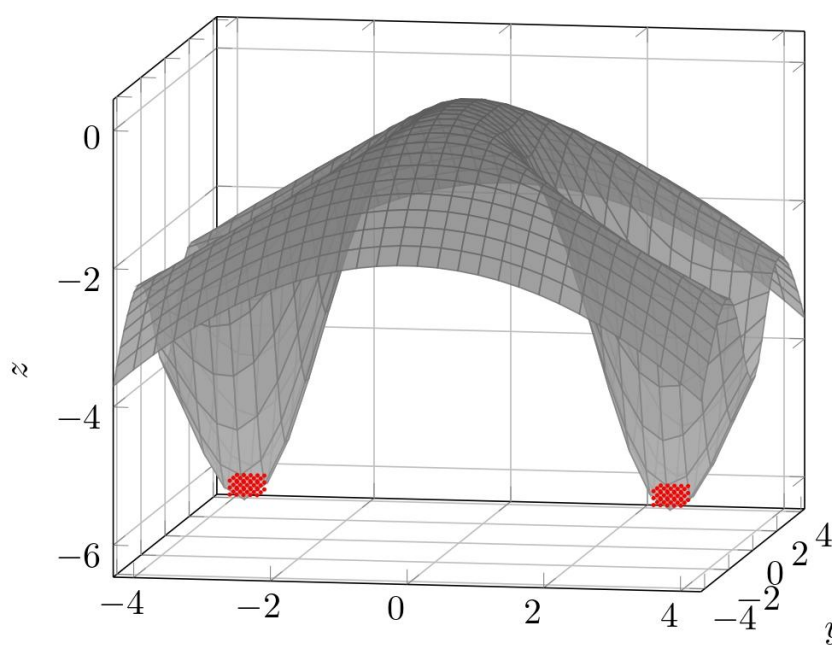


Figure 9.4: Diffuse irradiance from sky dome, circumsolar region, and horizon

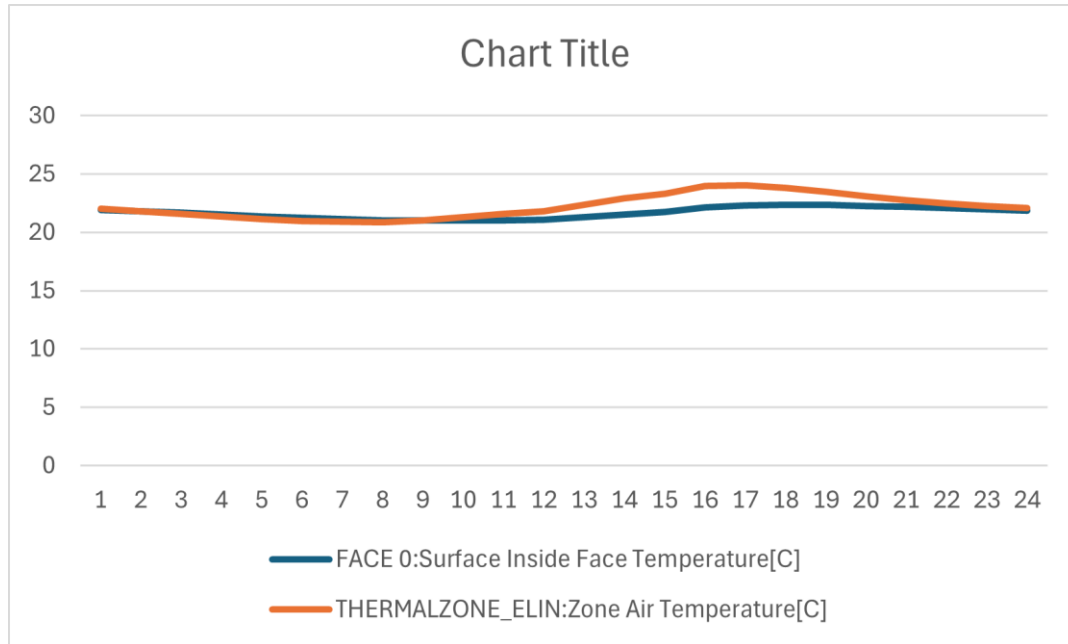
LaTeX, probably TikZ
From I. Beausoleil-Morrison (2021)

PGFPlots – Examples (Good for regular graphs)

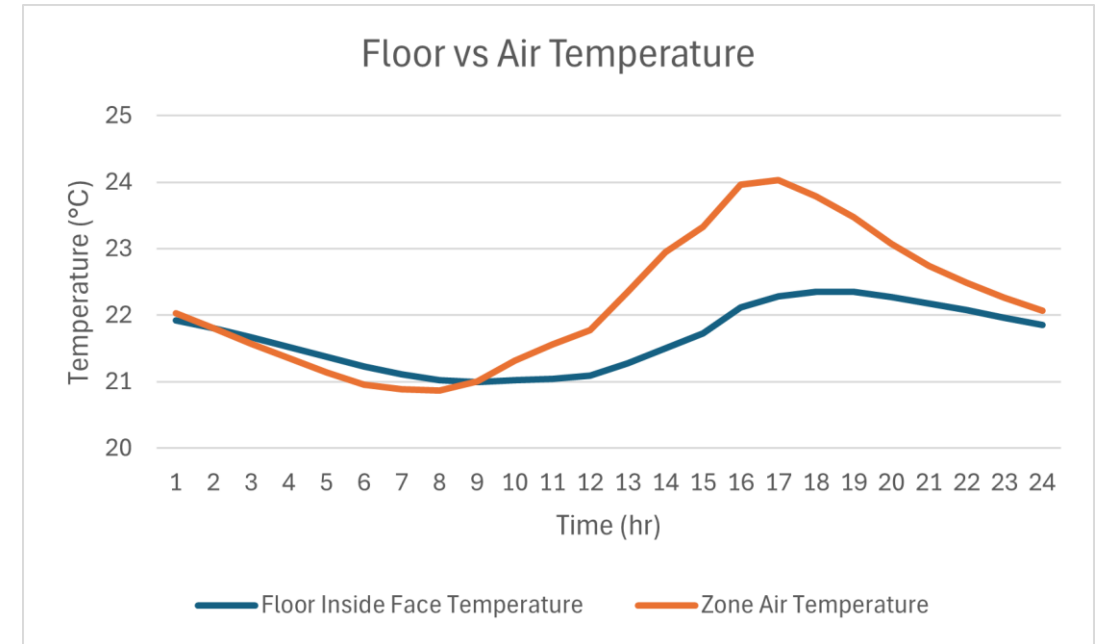


Source: Me!

Visualizations – Graph Fundamentals

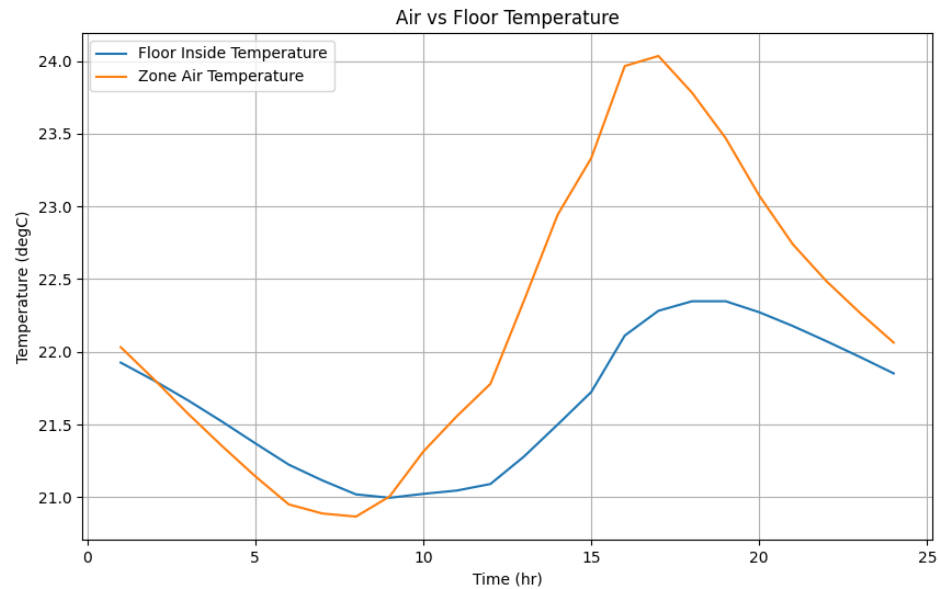


What's wrong with this graph?



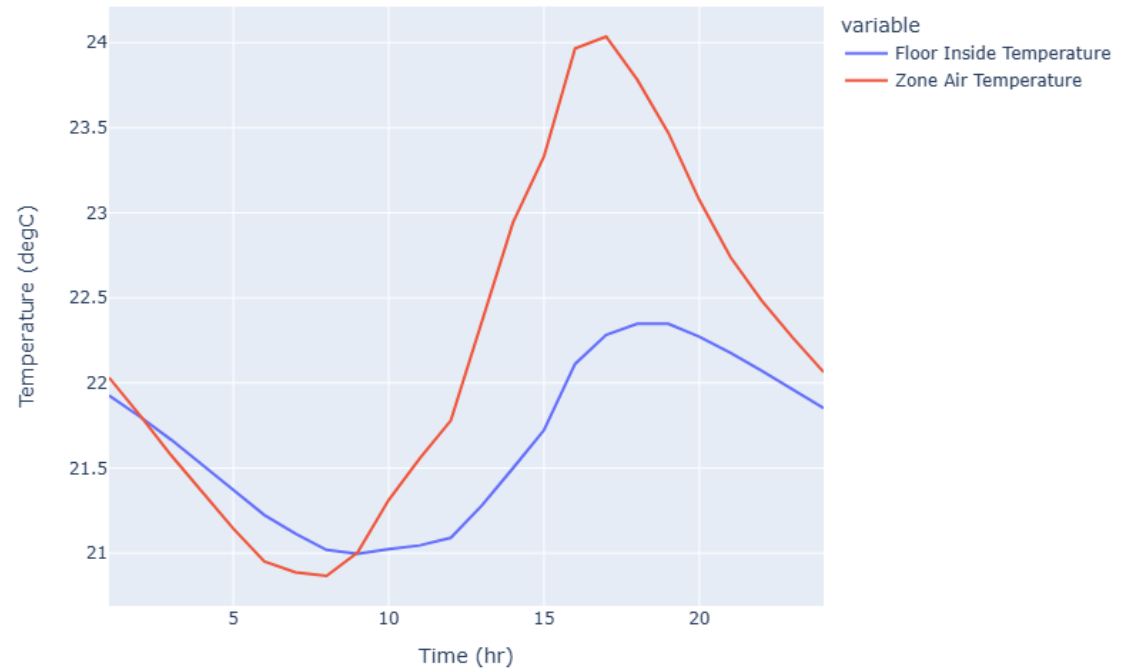
What makes this graph better?

The Graph We're Making Today



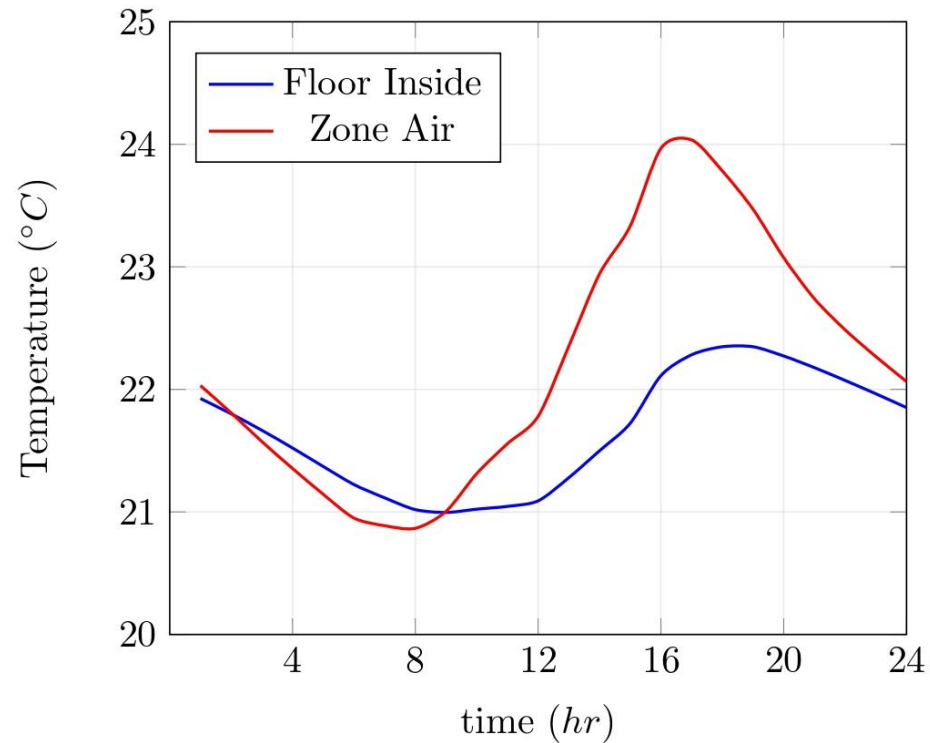
Python: Seaborn

Zone vs Floor Temperature

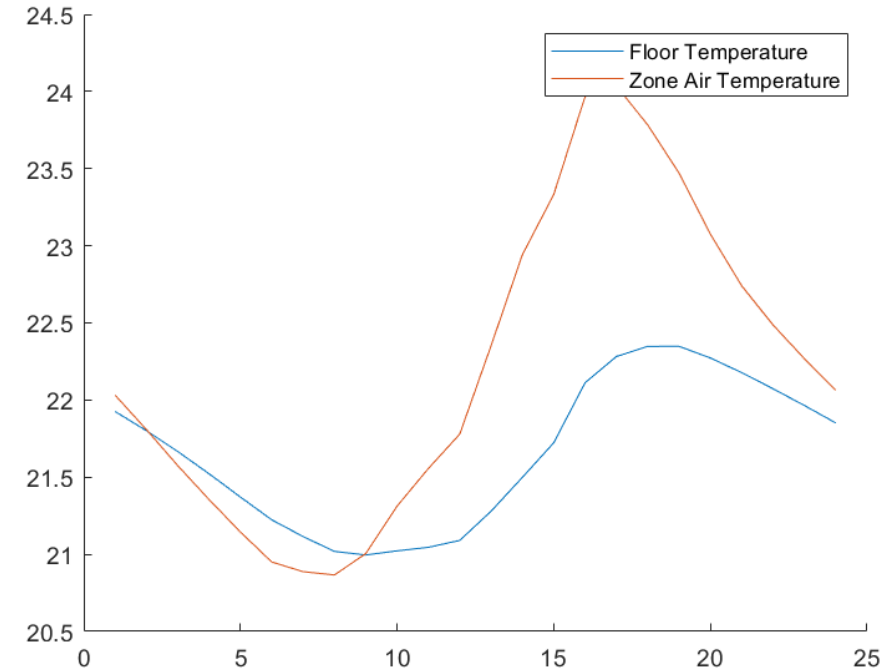


Python: Plotly

The Graph We're Making Today – Cont'd



LaTeX: PGFPlots – 16 lines of code



**MATLAB – 5 short lines of code
(+1 to fix label position)**

Both by me, both line counts excluding data import and preambles

Into the Abyss We Go!!



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Thank You!

Questions?

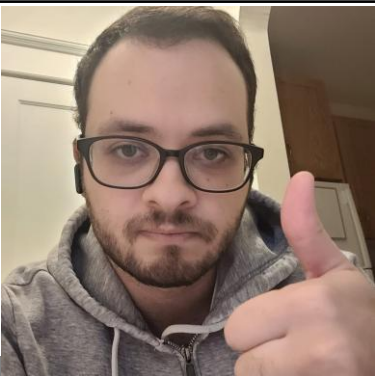
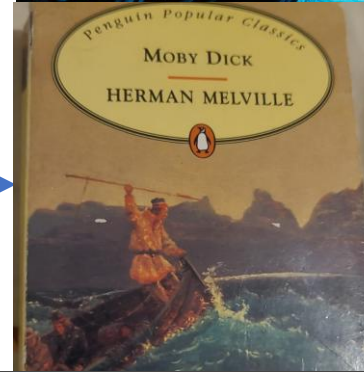
Introduction – *Moby-Dick*

Captain Ahab, and the
Pequod

Love/Hate Relationship



Love/Hate Relationship



Love/Hate Relationship

L^AT_EX